

SVI & MSX

SPECTRAVIDEO



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BANK SWITCHING*By, M. Gasiorowski*

Dear SAUG,

I have been able to adapt the Bank Switching program by Mr T. Colverd, printed in the Jan. 1986 newsletter, to run on a Spectravideo 328 with disk drives.

A simple patch in the program fixes the problem, which was basically due to the program over-writing reserved memory space, where the disk drive commands and buffers were stored.

To fix the program, simply change the following lines:

```
100 CLEAR 200,&HD500
110 DEFUSR=&HD500: DEFINIT I
120 FOR I=&HD500 TO &HD579
    and
150 DATA 22,5E,FE,E5,3E,C3,32,57,FF,21,34,D4,22,58,FF,ED
```

NOTE THESE CHANGES

The changed program has been working fine for a few days now, so I don't think there will be too many problems with it.

The memory above about &HD590 is reserved by Disk Basic, so any machine language routines should be saved before this limit. Writing to memory higher than this is not picked up by the computer, and so one can easily write over the disk drive routines and consequently destroy the disk directory if a file then happens to be saved. So it pays to be careful here!

Finally, for the members without a disassembler, I have included a disassembly of the disk drive version of the program, so that they may be able to work through it themselves, and find out how it was done. Nice piece of programming...

Mark Gasiorowski [92]
4 Ariel St,
Pennington SA 5013

P.S. How about publishing an updated members list?

P.P.S. And how about a 'contact' facility, where members can get in contact with others with the same interests?
I am sure there would be a lot of interest in this.

BANK SWITCHING: DISK VERSION SAUG, 1986 DISASSEMBLY

```

D500:225EFE LD (FE5E),HL
D503:E5 PUSH HL
D504:3EC3 LD A,C3
D506:3257FF LD (FF57),A
D509:2134D4 LD HL,D434
D50C:2258FF LD (FF58),HL
D50F:ED735CFE LD (FE5C),SP
D513:F3 DI
D514:3E0F LD A,0F
D516:D388 OUT (88),A
D518:DB90 IN A,(90)
D51A:3264FE LD (FE64),A
D51D:E6FD AND FD
D51F:D38C OUT (8C),A
D521:210080 LD HL,8000
D524:110000 LD DE,0000
D527:010080 LD BC,8000
D52A:EDB0 LDIR
D52C:3A64FE LD A,(FE64)
D52F:D38C OUT (8C),A
D531:FB EI
D532:E1 POP HL
D533:C9 RET
D534:FEC9 CP C9
D536:C0 RET NZ
D537:ED735CFE LD (FE5C),SP
D53B:DD2A03FA LD IX,(FA03)
D53F:23 INC HL
D540:225EFE LD (FE5E),HL
D543:F3 DI
D544:3E0F LD A,0F
D546:D388 OUT (88),A
D548:DB90 IN A,(90)
D54A:3264FE LD (FE64),A
D54D:E6FD AND FD
D54F:D38C OUT (8C),A
D551:210000 LD HL,0000
D554:110080 LD DE,8000
D557:4E LD C,(HL)
D558:EB EX DE,HL
D559:46 LD B,(HL)
D55A:71 LD (HL),C
D55B:EB EX DE,HL
D55C:70 LD (HL),B
D55D:23 INC HL
D55E:13 INC DE
D55F:7A LD A,D
D560:B3 OR E

```

```

D561:20F4 JR NZ,D557
D563:3A64FE LD A,(FE64)
D566:D38C OUT (8C),A
D568:FB EI
D569:ED7B5CFE LD SP,(FE5C)
D56D:CD5037 CALL 3750
D570:DD2203FA LD (FA03),IX
D574:2A5EFE LD HL,(FE5E)
D577:7E LD A,(HL)
D578:C1 POP BC
D579:C9 RET

```

PAGER

By, J. Robinson

This is a program which can store up to ten pages of on-screen information in the TEXT mode. At any time while BASIC is running, you may either hit SELECT or ^SELECT depending on whether you want to store or retrieve information.

ABOUT THE PROGRAM

Pager's memory space is the same as the PLAY statement's. It has been put there so as not to be over-written by variables and the like. A PLAY statement will cause the Pager function to be over-written: An action which may cause the computer to reset if Pager is "ON" at the time.

A SCREEN 1 or SCREEN 2 command will erase the pages you have stored. A SCREEN 0, though, will have no effect.

This program will not run under DISK BASIC (In theory, only: I have not tried it on a disk system).

Mr. L.A. Dunning must be congratulated on his series of articles on "Exploring Basic" it was information learnt from these which enabled me to write this program!

HOW TO USE

To store a screen : Hit ^SELECT. Choose a number between 0 and 9 and depress the appropriate key. The current contents of the screen will be stored under the number that you chose.

To display a screen : Hit SELECT : choose a numeric key according to what you want to display. It will appear on the screen.

IF you at the last moment decide that you do not want to store/retrieve a screen (i.e. after you have hit SELECT/^SELECT), then hit the space bar to resume normal functioning.

Pager (328/318)

by : J. Robinson

This Program may be entered using the 'INPUT' program from Newsletter 2 - 2 (NOV. 84.).

```

NP      10 REM | --      Pager      -- |
BM      20 REM | -- By J. Robinson -- |
FN      30 REM
HP      40 FOR I=&HFB0A0&&HFB09
BO      50 READ A$:POKE I,VAL("&h"+A$)
BO      60 NEXT
CO      70 POKE&HFF5A+2,&HFB
CM      80 POKE&HFF5A+1,&HAD
CA      90 POKE&HFF5A,&HC3
DD      100 POKE-2,0
AL      110 END
AN      120 DATA 3E,1,32,FE,FF,CD,BB,FB,0,3A,7B,FD,CB,4F,C2,23,FB,AF,32,FF,F
        F,0,C3,2B,FB
HP      130 DATA 3E,1,C3,1C,FB,3A,FF,FF,FE,1,CA,33,FB,C3,40,FB,CD,66,FB,1,C0
        ,3,11,0,0
FD      140 DATA EB,C3,49,FB,CD,66,FB,1,C0,3,11,0,0,EB,CD,47,37,DB,84,32,FF,
        FF,EB,CD,3C
EI      150 DATA 37,3A,FF,FF,D3,80,23,13,B,78,B1,C2,49,FB,C3,A2,FB,CD,3D,40,
        FE,20,CA,9C,FB
AB      160 DATA D6,30,32,FF,FF,11,C0,3,21,C4,31,1,9,0,3A,FF,FF,B9,CA,92,FB,
        B7,ED,52,78
FF      170 DATA B1,B,C2,7C,FB,CD,BE,40,C3,66,FB,FE,0,CA,98,FB,C9,21,84,35,C
        9,E1,21,A2,FB
EP      180 DATA E5,C9,AF,32,FE,FF,2A,CA,FB,C3,C3,FB,C9,DB,99,FE,EF,C0,3A,FE
        ,FF,FE,0,CA,A
BG      190 DATA FB,C9,2A,1A,FA,0,22,CA,FB,C9,22,1A,FA,22,1C,FA,C9
CI      200 REM
BC      210 REM      Memory Map
CK      220 REM
BE      230 REM FFFE H : A zero must be placed here for the routine to work.
BA      240 REM FF5A H : The vector used to jump to the routine.
BD      250 REM FBAD H : The entry point of the routine.
BH      260 REM FB0A H - FBC9 H : The routine.
CP      270 REM
CF      280 REM To switch the Pager function on or off, use:
CO      290 REM POKE &HFF5A , &HC9 for OFF
HI      300 REM POKE &HFF5A , &HC3 for ON
CI      310 REM
GB      320 REM You may dispose of this program after it has been RUN.
END

```

LETTERS TO ANSWER

Dear S.A.U.G.

I am looking for more information regarding cursor location using ESC "Y" as referred to in the CP/M users guide Page 7.

I.E. ESC Y - Locate cursor, X and Y address follow

Do the X,Y addresses refer to column and row? Which corner is the Origin.

The reason for this info is that I am trying to write an assembly language program to run under CP/M that requires Cursor Location.

G. Perry
2 Royston Place,
Mt. Helena
West Australia 6555

The Following letters need you the members to help them.

Dear S.A.U.G.

I now have the 1985 Year Book, & issues 3-3, & 3-5/6 of the Mag, and find them excellent. I would however like to obtain copies of the 84 Year Book & issues 3-1, 3-2 & 3-4, Especially for the "Exploring Basic" Sections for the Kids.

Is there anyone in Melbourne (Preferably Eastern Suburbs) where I could borrow them to copy?

Also you said the SV 318 & M.S.X. tape formats are not compatible, but what are the formats? How are they Different?

Ian Davidson,
19 Warnes Road,
Mitcham VIC. 3132

TIME COMMAND FOR 328/318

By, B. Curwood

```

; New commands

; This program is a demo of how to
; add new commands to BASIC by
; prefacing them with the CMD
; command. In this example the
; following 4 commands are checked
; with a jump to 4 subroutines as
; appropriate. If there are any
; errors then the routine exits to
; the SYNTAX ERROR in XXX routine

```

```

;CMD TIME SET (or CMDTIMESSET)
;CMD TIME ON (or as above )
;CMD TIME OFF (" " " ")
;CMD TIME PRINT

```

```

;Note the following points

```

```

;1. If the command is accepted the
;HL pair should be left pointing
;to the next byte after the CMD
;(whatever). Thus after the CMD &
;the TIME XXXX is accepted the HL
;pair is pushed, saving the
;address of the NEXT command on
;the stack & after the required
;subroutine is performed it is POP
;-ed and the return address to the
;CMD routine is removed by a dummy
;POP AF.

```

```

;2.The easiest way to implement
;a new command is to use one of
;the existing reserved words as
;they are tokenised & thus only
;1 or 2 bytes need checking.
;However, it is possible to make a
;completely new command word but
;it means that each letter needs
;checking. (Unless some of them are
;a reserved word, see the TO in
;the middle of MONITOR)

```

```

;3.You could also use the MON word
;by using the hook at FF84H.
;For example a new command of
;MONITOR would call your routine
;(via the hook) with the HL pair
;pointing to the start of ITOR.
;See the last example

```

```

;4.To use the CMD routine you must
;place a jump to your routine at
;the CMD hook, FF9CH.

```

```

;5. After the call to CMD (or MON)
;HL is pointing to the next non-
;space letter of your command, thus
;if you only have a 2 word command
;e.g CMD TIME then it doesnt
;matter how many spaces in between

```

```

;them but if you have 3 (or more!)
;then it is your job to check for
;a space & jump over if nessesary.
;See SPACE1

```

```

;6.The print routine is only used
;to demonstrate the result of a
;command. It positions the cursor
;at the 2 locations before each
;message (i.e X,Y) provided they
;are not both zero in which case
;the message is printed at the
;last known cursor position.

```

```

;7. The RST 10H checks that the
;byte after the new command is
;either a ':' separating it from
;the next command or the end of
;the line, in either case it
;will return with the Z flag set
;if OK.

```

```

;To use this example program first
;ASSEMBLE it and then use the M
;command (if you are using ZEN) to
;change the CMD hook to C3 1D 90 &
;change the MON hook to C3 75 90.

```

```
*****
;*****
;****          CMD TIME SET, DISPLAY, ON, OFF          ****
;****          ****
;****          by          ****
;****          Bob Curwood          ****
;****          Technical Officer          ****
;****          Christchurch Spectravideo          ****
;****          Users Group          ****
;****          ****
;*****
```

```
ORG 9000H
LOAD 9000H
```

```
CLS: EQU 03777H
CHPUT: EQU 00018H
POSIT: EQU 0393EH
ERROR: EQU 00907H
CSRY: EQU 0FA03H
CMDHK: EQU 0FF9CH
```

```
TIME: EQU 0EFH
SET: EQU 0D2H
ON: EQU 095H
OFF: EQU 0EBH
PRINT: EQU 091H
```

```
TEST:          CALL ALTERHK
              JP 0A003H          ;Jump to ZEN for now
;Delete above 2 lines after de-bugging
ALTERHK:        LD HL,CMDHK
              LD A,0C3H
              LD DE,CMD
              LD (HL),A
              INC HL
              LD (HL),E
              INC HL
              LD (HL),D
```



```

                                RET                                ;Return to BASIC

CMD:      LD A, (HL)                                ;Jump here from CMD
          CP TIME
          JR NZ,EXIT3
SPACE:    INC HL                                ;Error exit
          LD A, (HL)
          CP " "
          JR Z,SPACE                                ;Jump over spaces
          PUSH HL                                ;Address of next command
          CP SET
          JR Z,SETIME
          CP ON
          JR Z,TIMEON
          CP OFF
          JR Z,TIMOFF
          CP PRINT
          JR Z,PRTIME
          JR EXIT3                                ;None of these!

SETIME:   LD DE,MS1
          LD HL, (CSRY)
          CALL PRINT1
          CALL POSIT
          JR EXIT4                                ;OK Exit

TIMEON:   LD DE,MS2
          LD HL, (CSRY)
          CALL PRINT1
          CALL POSIT
          JR EXIT4

TIMOFF:   LD DE,MS3
          LD HL, (CSRY)
          CALL PRINT1
          CALL POSIT
          JR EXIT4

PRTIME:   LD DE,MS4
          LD HL, (CSRY)
          CALL PRINT1
          CALL POSIT
          JR EXIT4

EXIT3:    LD E,2
          JP ERROR                                ;Syntax error

EXIT4:    POP HL
          RST 10H
          JR NZ,EXIT3
          POP AF
          RET                                ;Check for : or 00 byte

PRINT1:   PUSH HL
          EX DE,HL
          LD D, (HL)
          INC HL
          LD E, (HL)
          LD A,D
          OR E
          JR Z,LOOP2

```

```
                EX DE,HL
                CALL POSIT
                EX DE,HL
LOOP2:          INC HL
                LD A,(HL)
                AND A
                JR Z,EXIT2
                CALL CHPUT
                JP LOOP2
EXIT2:          POP HL
                RET

MS1: DB 32,1,"SET TIME",0
MS2: DB 32,1,"TIME ON ",0
MS3: DB 32,1,"TIME OFF",0
MS4: DB 32,1,"12:45:18",0
END
```

MONITOR PROBLEM

"Dear S.A.U.G.

My problem concerns the horizontal positioning video output to my TV monitor. The display is left to the extent that on 40 column display the first column is cut off. My TV is set up correctly for a transmit pattern, and I have been unsuccessful in trying to adjust around the VDP chip.

All other SV's I've seen suffer the same problem. Can you help me with a hardware fix ? "

Bruce Goodman
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MAWSON
ACT 2607

Telephone: (062) 86 4002 (AH)